

# Oliver Zhang

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## Summary

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I'm a versatile mechanical engineer with extensive experience designing and building complex electro-mechanical systems. I thrive in highly collaborative interdisciplinary teams, and I love finding elegant solutions that bring innovative ideas to life.

I'm an Australian citizen.

## Education

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**Carnegie Mellon University**, BS in Mechanical Engineering with an additional major in Robotics with University Honors – Pittsburgh, USA      Aug 2015 – May 2019

- Completed a wide range of course work from mechanical engineering fundamentals to computer vision, artificial intelligence, and systems engineering.
- Worked with cross-disciplinary teams on multiple projects, including building an amphibious vehicle, a self-navigating drone, and an autonomous delivery robot.

## Experience

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**Senior Mechanical Engineer**, PA Consulting – San Francisco, USA      July 2019 – present

Collaborating with diverse engineers and designers to find the path through complex challenges and turn inventive ideas into innovative products.

- Delivering projects across many different industries, from medical robotics to industrial food production.
- Working through the entire product development process, from concept to manufacturing.
- Collaborating with a wide range of clients from agile tech startups to established global leaders.
- Brainstorming ideas and developing concepts to find the best path forward.
- Building prototypes at the right fidelity to rapidly drive to the optimal design.
- Architecting complex systems by breaking them down into key interactions and applying first principles analysis and modelling.
- Coordinating highly interdisciplinary teams spread across multiple geographies, with electrical engineers, software engineers, and industrial designers.
- Growing the team's capabilities through weekly forums and training sessions, covering topics like analysis, sketching, or programming.
- The team's go-to expert for robotics, systems engineering, and analysis.

**Project Engineer Intern**, Autel Robotics – Pittsburgh, USA      May 2017 – Aug 2017

Implemented tracking and gesture recognition algorithms for embedded execution on a quadcopter.

## Skills

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**Programming:** Python, C/C++, MATLAB, ROS, Unity

**Mechanical Engineering:** Technology development and prototyping, Analysis of complex systems, Electromechanical systems, Design for manufacture, SolidWorks, GD&T, Tolerance stack analysis

**Systems Engineering:** Requirements development, System architecture design, Risk management

**Languages:** English (native speaker), Mandarin (native speaker)

## Patents

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**US11058204B2**      July 2021

Automated total nail care systems, devices, and methods

## Select Projects

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### Novel Electromechanical Machine

Aug 2025 – Feb 2026

Developing a consumer facing electromechanical machine for a global technology company.

- Owned design of an Alpha prototype system with 7 actuated degrees of freedom.
- Designed complex motion systems in heavily space constrained and finish constrained environment.
- Balanced strict industrial design intent with reliability, while meeting aggressive schedule.
- Interfaced with a diverse team of electrical engineers, software engineers, and industrial designers.
- Led prototype build effort, coordinating integration of PLC electronics, wiring of motors and sensors, and assembly of complex mechanical subsystems.
- Final prototype's reliability and quality exceeded expectations and allowed client to fully de-risk intended user experience with company executives.

### Novel Cable Splicing Machine

Aug 2024 – Aug 2025

Developing an underground cable splicing machine that automates a complex process typically done by a trained technician.

- Built complex system requirements drawn from multiple stakeholders, and managed key risks.
- Designed system architecture to maximise reliability and meet stringent weight and space constraints.
- Developed multiple electromechanical mechanisms that each perform dexterous operations that previously required highly coordinated human actions.

### Novel Smart Fitness Machine

Aug 2022 – Aug 2024

Developed a novel fitness machine that is controllable wirelessly and can report data from multiple integrated sensors.

- Implemented and refined controls algorithms to realistically simulate inertia with an electric motor.
- Built dynamometer rig with software that simulates human motion and displays metrics in real-time.
- Developed system requirements, managing traceability and designing test plans.
- Led verification and validation of the prototypes across multiple locations globally.
- Coordinated a highly cross-functional team with industrial designers, electrical engineers, firmware engineers, and mechanical engineers.
- Delivered high quality prototypes that demonstrated best-in-class user experience and performance, standing up to rigorous testing by world-class athletes.

### Novel Subsystem Prototype

Sept 2020 – Apr 2021

Developed a proof-of-principle prototype to de-risk a key robotic subsystem.

- Developed a complex analytical model of the behavior of elastic materials fed from a spool.
- Experimented with medical grade extruded tube materials, coatings, and cross-sectional profiles to generate desired properties such as bending stiffness, resilience, creep, and friction.
- Designed and built a robotic system that process various sensor inputs to drive a custom control algorithm. The system uses clinical trial data to simulate real user motion, allowing high fidelity testing of expected behavior.

### Evolv Express® Threat Detection System

Nov 2019 – Sept 2020

Developed manufacturable prototype in support of client electronics, addressing challenging structural, materials, mobility, and environmental requirements.

- Architected mechanical structure to improve stiffness, assemblability, and serviceability.
- Converted fibreglass sheet-based design into structural foam/RIM/plastic extrusions.
- Performed comprehensive Six sigma tolerance stack analysis and implemented GD&T tolerance controls.
- Managed complex assembly with master modelling techniques.
- Final prototypes secured customers and funding, eventually leading to public listing of client's company.